Additional Analysis of the Dataset

July 7, 2026

Here, we show some further analyses of the dataset and compare it to the "critical" dataset. This document provides further suggestions on how one can analyze the use of the chatbot and what insights can be drawn from the analyses.

1 Word Count

First, we examine whether a simple prompt property—the number of words—differs across classes. Figure 1 shows boxplots of word counts for each topic. Medians are generally similar, though *other* and *critical* have lower counts and narrower ranges, with minimums and maximums close together. In contrast, *interests*, *general*, and somewhat *matching* show much wider ranges, lacking a "typical" word count. The remaining topics have moderately narrow, comparable distributions. Overall, word count varies across topics, but no clear patterns emerge.

For the different prompt types[1], no clear patterns emerges either. The median for type *answer* is slightly higher (median = 11), while *follow-up request* is lower (median = 8.5) compared to the other types, though based on a few cases. Distributions for *statement* (SD = 21.80) and *answer* (SD = 15.53) are more spread out than the others.

As can be seen in Figure 2, uncertain prompts tend to be slightly longer with broader distributions, reflecting the extra phrases like *"I guess"* or *"But I'm not sure"* used to express uncertainty.

[1]A figure showing the boxplots for the different types is available in our online repository.

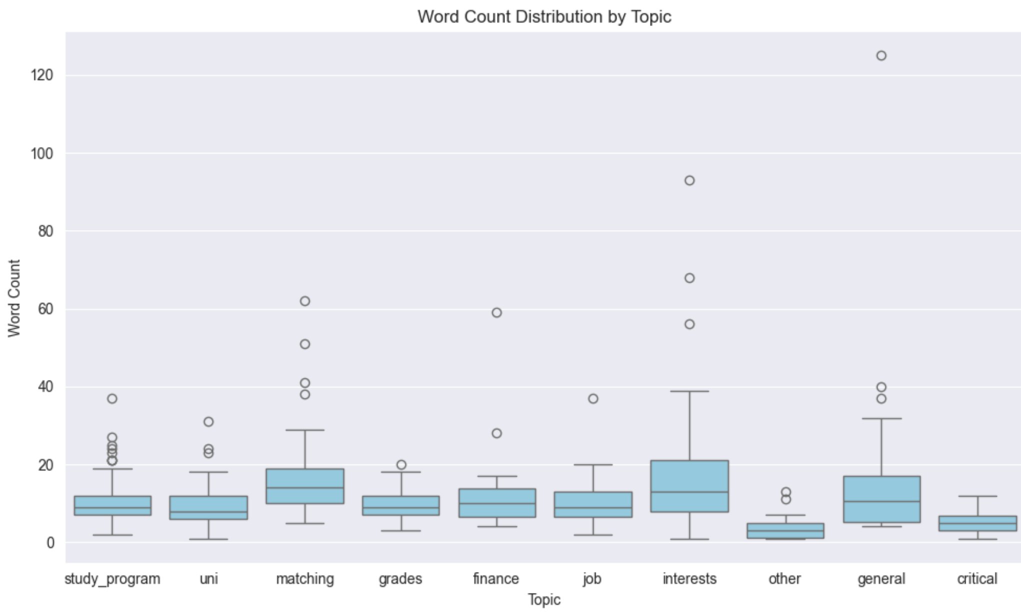

Figure 1: The boxplots representing the distribution of the word counts separately for each topic.

All in all, we can see some differences in word count across the classes, but there are no absolutely clear patterns. Some classes have a much narrower distribution than others, indicating that the way the prompts are phrased may be quite similar within this class.

2 Features

Next, we examine the features extracted by the SAE. To compare certain versus uncertain prompts, we first identified features present in at least two certain prompts and checked whether they occurred in any uncertain prompts, then repeated this for the uncertain prompts. As expected, more features appear multiple times in certain prompts but never in uncertain ones: 315 features are absent in uncertain prompts, versus 20 that are absent in certain prompts. Among features unique to uncertain prompts, some are intuitively meaningful, such as *unsure*, *not knowing what or how next*, and *no followed by ideas or control.*

Proceeding like this for the critical prompts versus the non-critical prompts

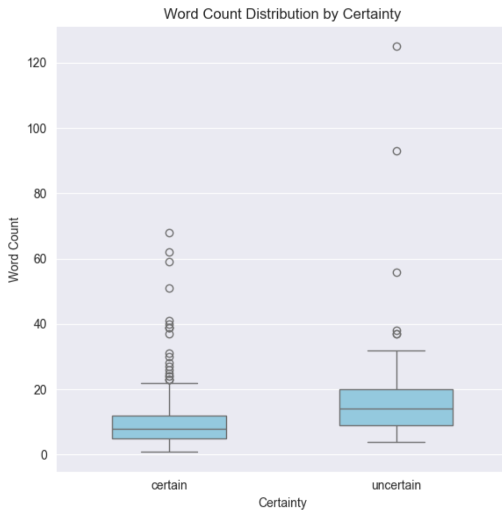

Figure 2: The boxplots representing the distribution of the word counts separately for certain and uncertain prompts.

in the main dataset, we, again, obviously see that more features are never seen in the critical prompts, of which there are much fewer: 4 features that appear more than once in critical prompts are never seen in non-critical prompts, versus 487 the other way around. Looking at the four features in the critical prompts that are never seen in the non-critical prompts, they mostly appear sensible, although they do not contain inherently problematic information: *tone*, *names of people*, and *"You are" statements* can all easily be imagined as features of problematic prompts.

These first observational findings indicate that the features extract meaningful information that can be used to identify the different classes.

3 Comparison of the Critical Prompts

We also compare the critical prompts in the main dataset with those in the critical dataset to see whether there are signs of structural differences. 14 features appear more than once in the main dataset's critical prompts but never in the critical one, and 69 for the opposing constellation. As the critical dataset contains more prompts, again, this asymmetry in size is not entirely surprising.

Looking at the features absent from the critical dataset, we observe that they often relate to specific people or the tone of the prompt. Moreover, some prompts concern gambling and, accordingly, there are repeated features concerning this topic.

Looking at features not present in the main dataset's critical prompts, many features are rather general and do not seem to inherently map to problematic aspects (e.g., *want to*, *months and dates*, *with a descriptive noun*). The features that are more obviously connected to problematic aspects in the prompts are often political or discriminatory ones, e.g., *Palestinian health*, *war and conflicts in regions*, or *women are or women's*. Repeated features in both datasets are profanities and insults.

Generally, we may observe that the prompts in the main dataset, gathered from current high school students, appear less political or discriminatory than we might expect. Instead, many of the prompts include simple profanities. However, looking at the prompts in detail, there are also more nuanced ones that are critical for other reasons. For example, some prompts ask for particularly easy study programs or whether professors at specific institutions are nice. Such questions should not be answered directly but rather handled with care.

All in all, we may conclude that there are differences between the critical prompts in the main dataset and those in the critical dataset. It is hard to say, however, whether these differences also hold once the chatbot becomes publicly available. Certainly, we still need to monitor the prompts, and our critical dataset may be valuable for training a classifier to identify the critical prompts.

4 Additional Figures

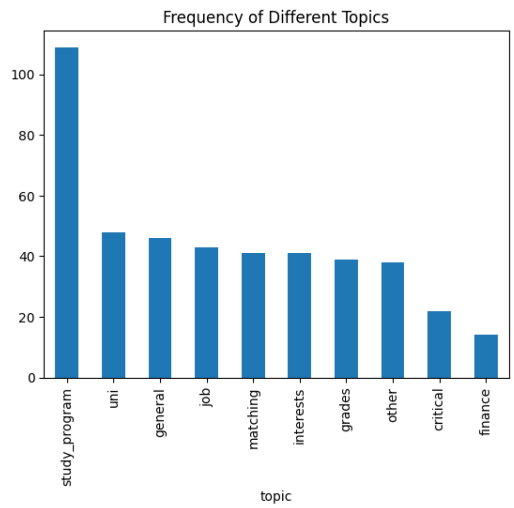

Figure 3: Frequency of different topics of prompts.

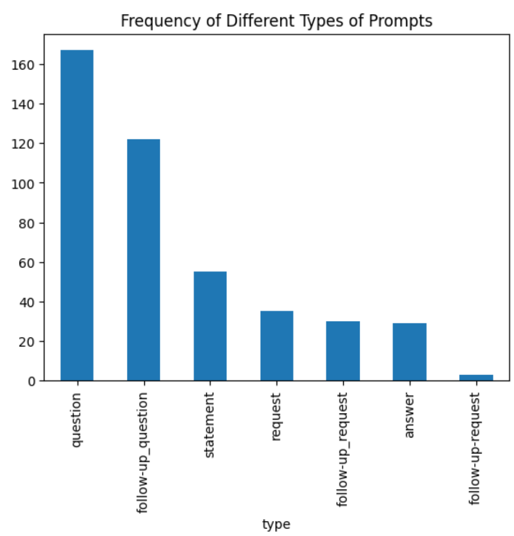

Figure 4: Frequency of different types of prompts.

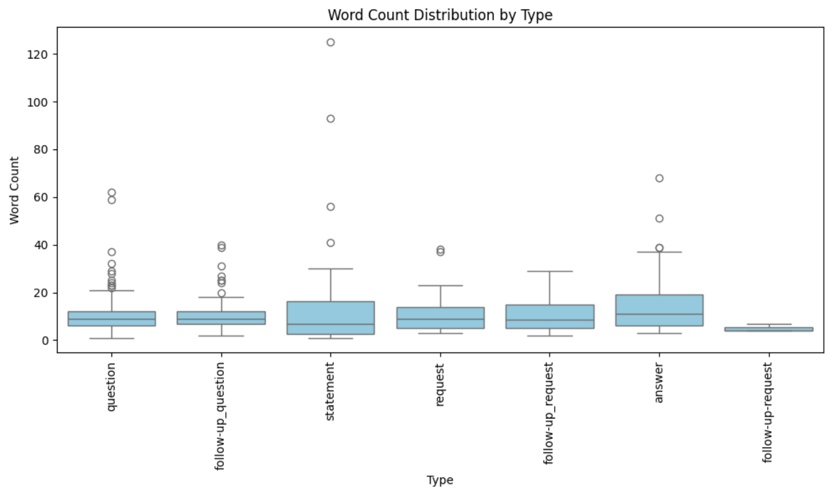

Figure 5: Word count distribution by type.

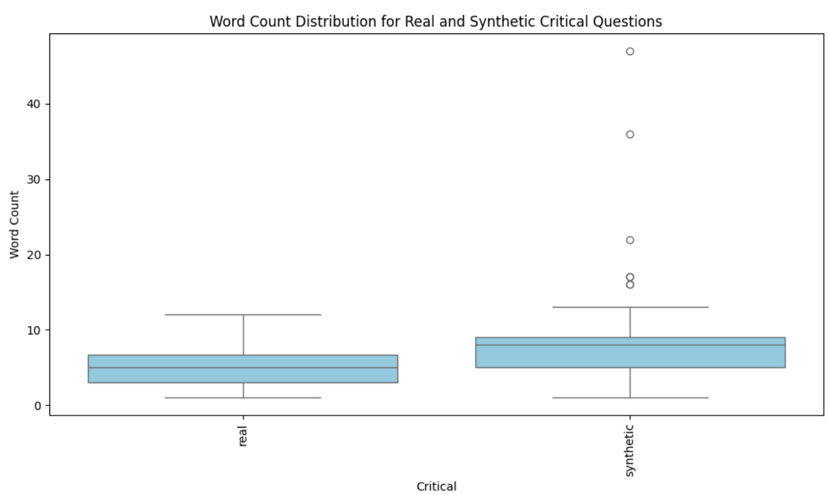

Figure 6: Critical prompt word count grouped by datasets.